PiNNTORP LLC

Online Gambling Platform

Phase 2: Minimum Viable Product Submission

CSCI 2040U - Software Design and Analysis

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1. MVP Functionality & Overview

1.1 Summary of Iteration 1

Iteration 1 ran from February 24 to March 9, 2026 (14 working days). All planned iteration 1 features were successfully implemented and delivered, having completed 26 of 26 developer-days of tracked implementation work at a velocity of 26 developer days per iteration. The MVP itself, is a fully browser runnable vanilla JavaScript web application (HTML/CSS/JS) with no external framework dependencies. The persistence is handled via localStorage, and the Java back end (WebSocket server) was initiated and scaffolded, but isnt required to run the current MVP.

The following core capabilities have been implemented:

- Two functional gambling games: a Slot Machine and a Dice Roll game, both with correct win/loss/profit logic and initial bet mechanics, defaulting to a $5 bet.
- A Statistics Display displaying the amount of games played, wins, win/loss ratio, and profit per user so the users can track their performance quickly.
- A Friends List system supporting add, view, and remove operations with localStorage persistence.
- A modular JavaScript architecture (i.e. state.js, storage.js, game.js, friends.js, stats.js, etc) that separates the core functions from the displays.
- A Java WebSocket server skeleton under pinn-api/ that will help in the future real time features.

And the following features are deferred to other iterations and are not in this MVP:

- Recommendation algorithm (Iteration 2)
- Account deletion (Iteration 3)
- App UI Polish (Iteration 2/3)

1.2 Implemented User Stories

User Story	Priority	Effort	Status
As a user, I want functional game mechanics so I can use the product.	Must Have	14 days	Implemented
As a user, I want to see my statistics to make decisions.	Must Have	4 days	Implemented
As a user, I want my win/loss ratio so I can brag.	Must Have	4 days	Implemented
As a user, I want to add friends to see who is active.	Should Have	4 days	Implemented
As a user, I want game recommendations so I don't get bored.	Should Have	10 days	Deferred to iteration 2
As a user, I want to delete my account for data security.	Should Have	6 days	Deferred to iteration 3
As a user, I want pretty aesthetics and flashing win colours.	Nice to Have	6 days	Partial / Later

1.3 Known Gaps & Limitations

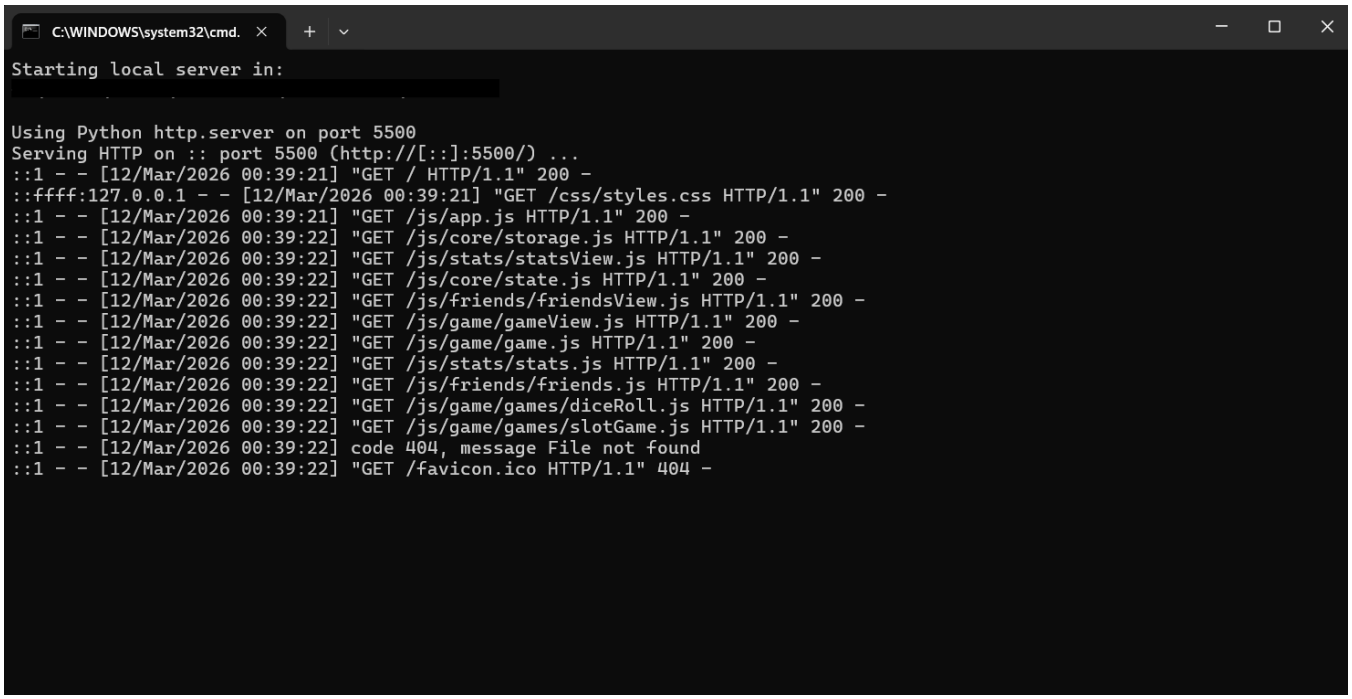
- The recommendation algorithm is not yet implemented, which is the primary focus of Iteration 2.
- The account deletion system is deferred to Iteration 3.
- Persistence is currently stored locally on the browser (in localStorage), the server side persistence will be implemented once the Java back end has been fully wired.
- The Java WebSocket server (under pinn-api/) is scaffolded but not integrated at all with the front end in this MVP.
- The bet amount is hardcoded at $5 per round in the games, implementing dynamic betting will be a future task to enhance the app.
- Only two of the planned three gambling games are implemented (Slots and Dice Roll). A third game is planned before final submission, as well as a revamp of the Dice Roll game and improvements to the Slots game.

2. MVP Walkthrough

2.1 Setup & Launch

The application is a static web app and requires no build step. To run the MVP locally:

- Clone the repository from GitHub and navigate to the project root.
- Double-click start-server.bat (on Windows), or run python -m http.server 5500 in the project root.
- Open http://localhost:5500 in any browser.

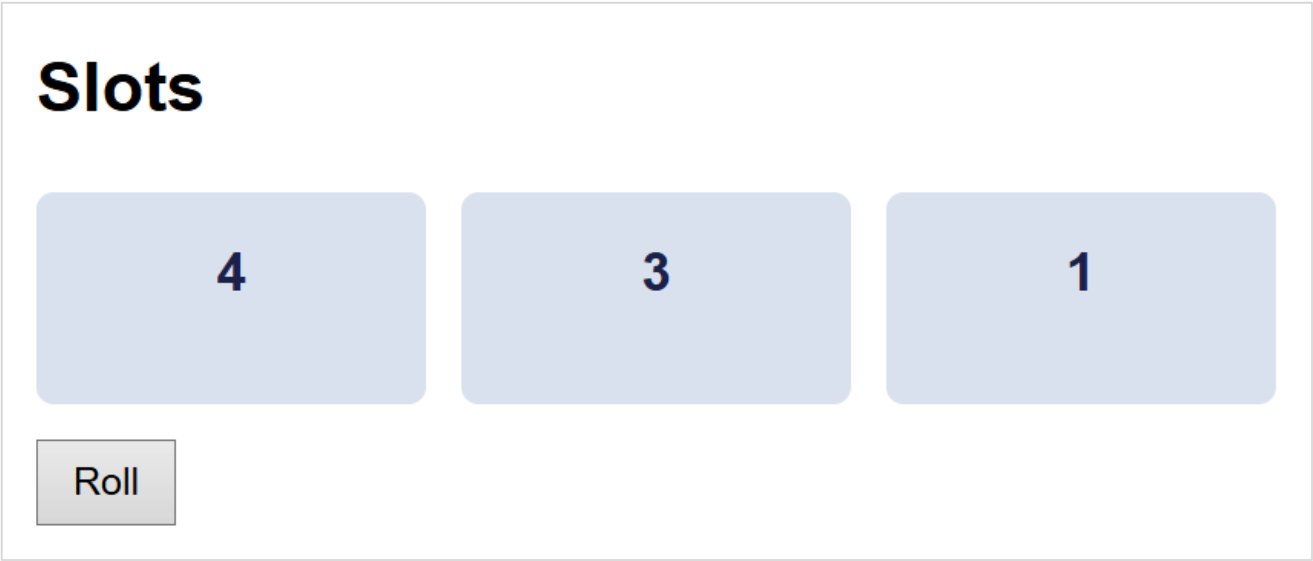


2.2 Slot Machine Game

From the main page, navigate to the Games column on the right and you will see the “Slots” cell. When you press on the “Roll” button, you will start the game and spin. Each spin generates three random numbers (1 through 7), and the outcome rules are:

- 3-of-3 match: the user wins (bet * the matched number), and the profit of the user increases accordingly.
- 2-of-3 match: the user wins a $10 bonus.
- No match: the user loses the $5 bet, and the profit of the user decreases accordingly.

After each spin, the result is immediately displayed and the statistics display updates with the new game amount played, win/loss count, and cumulative profit.

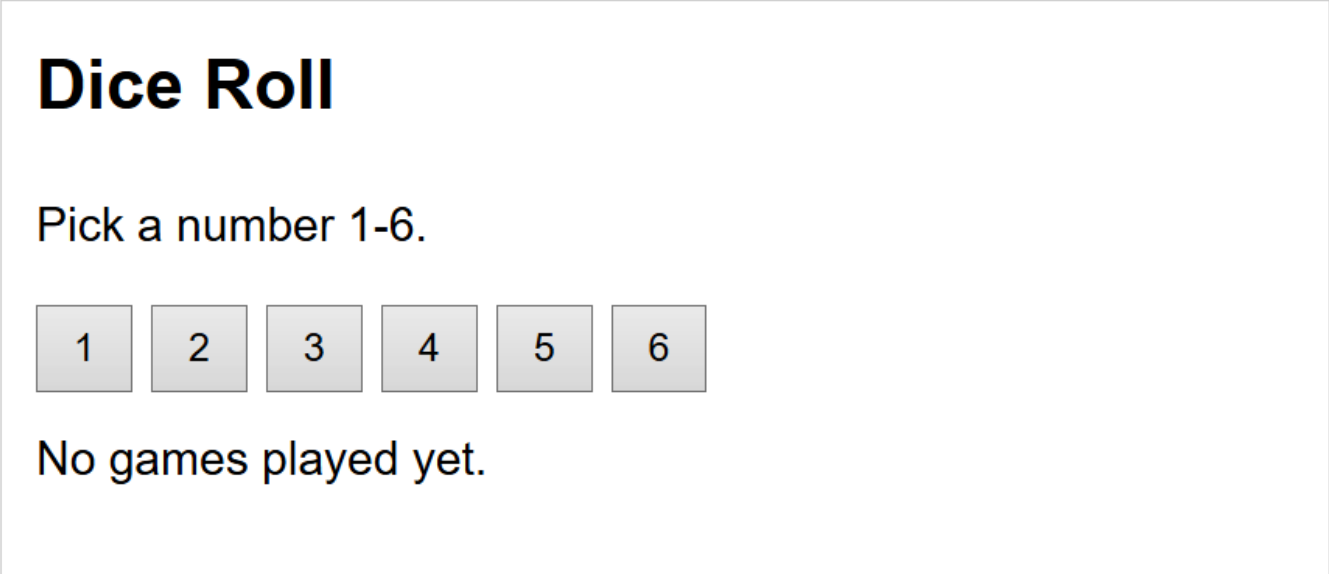


2.3 Dice Roll Game

From the games column, select “Dice Roll.” The user enters a guess (by clicking buttons 1 through 6), and a random die value is generated:

- Correct guess: user wins $10 (profit +10).
- Incorrect guess: user loses $5 (profit −5).

Results and statistics update instantly after each roll.



2.4 Statistics Dashboard

The statistics view shows the user's current totals: their total games played, total wins, win/loss ratio, and cumulative profit. All of the values persist across page refreshes via localStorage. The displays re-renders automatically after each game round ends with the updated stats.

2.5 Friends List

The friends section allows the user to:

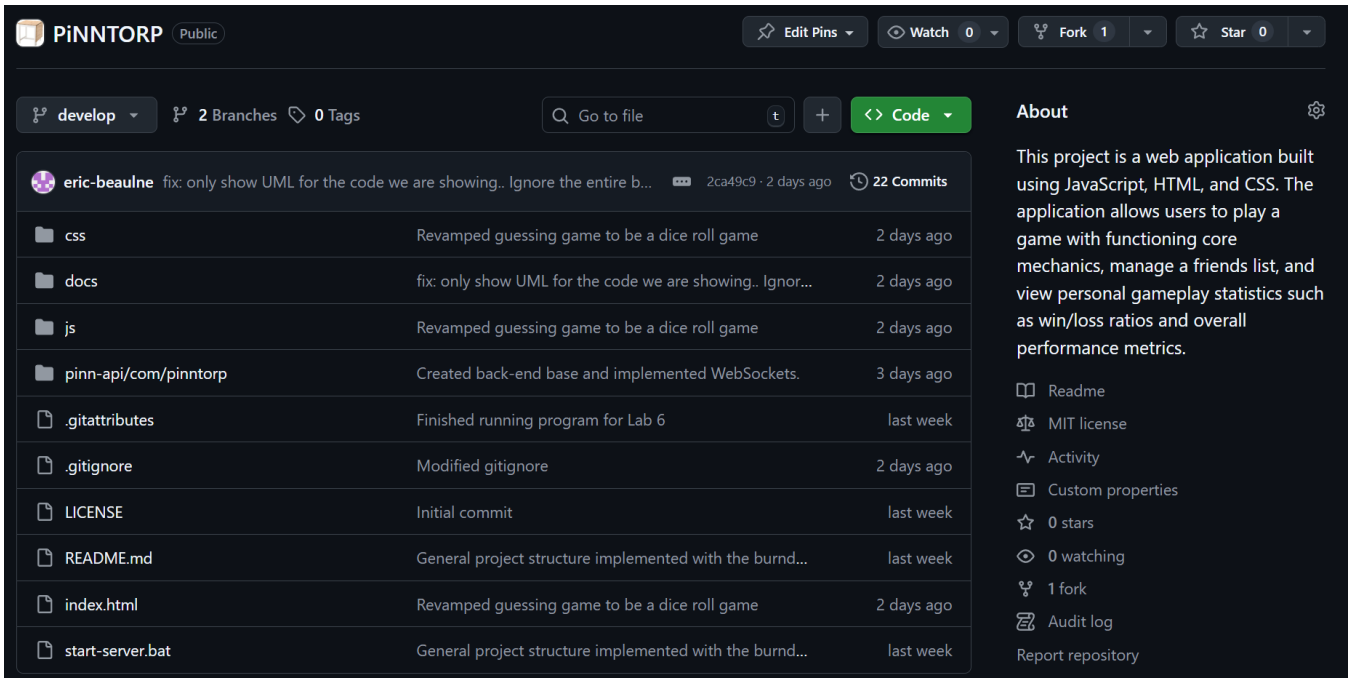
- Add a friend by entering their username in the field and clicking on “Add”.
- View the full list of their current friends.
- Remove a friend from the list by locating their name on the list, and clicking “Delete”.

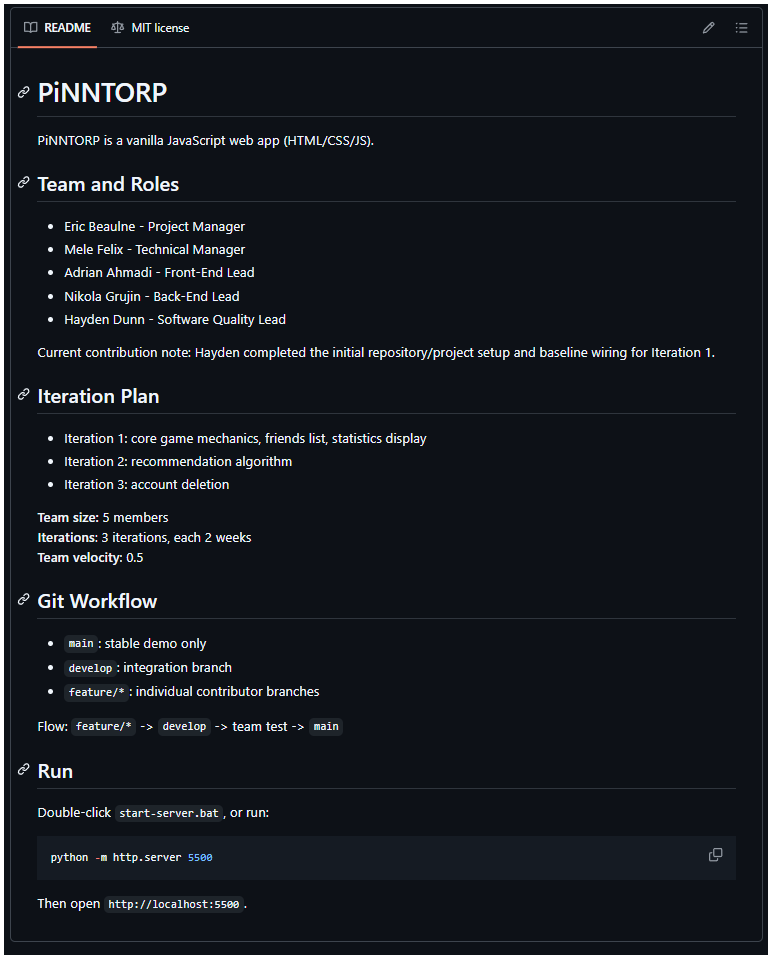
Friend data persists in localStorage, and is available immediately on next load.

3. Repository / Project Board

3.1 Repository Organization

The GitHub repository follows this structure:





README MIT license

PiNNTORP

PiNNTORP is a vanilla JavaScript web app (HTML/CSS/JS).

Team and Roles

- Eric Beaulne - Project Manager
- Mele Felix - Technical Manager
- Adrian Ahmadi - Front-End Lead
- Nikola Grujin - Back-End Lead
- Hayden Dunn - Software Quality Lead

Current contribution note: Hayden completed the initial repository/project setup and baseline wiring for Iteration 1.

Iteration Plan

- Iteration 1: core game mechanics, friends list, statistics display
- Iteration 2: recommendation algorithm
- Iteration 3: account deletion

Team size: 5 members
Iterations: 3 iterations, each 2 weeks
Team velocity: 0.5

Git Workflow

- `main` : stable demo only
- `develop` : integration branch
- `feature/*` : individual contributor branches

Flow: `feature/*` -> `develop` -> team test -> `main`

Run

Double-click `start-server.bat` , or run:

```
python -m http.server 5500
```

Then open `http://localhost:5500` .

Path / Branch	Purpose
main	The stable demo branch
develop	The integration/development branch with all feature branches merging here first
feature/*	The individual contributor branches per divided task
css/styles.css	The global stylesheet
js/core/	state.js (global state), storage.js (localStorage wrapper)
js/game/	game.js, gameView.js, games/slotGame.js, games/diceRoll.js
js/friends/	friends.js (logic), friendsView.js (UI)
js/stats/	stats.js (logic), statsView.js (UI)
pinn-api/	Java WebSocket server skeleton
docs/	Iteration 1 task board, burndown chart, UML diagrams, retrospective
README.md	Project overview, team roles, run instructions, Git workflow

3.2 Kanban Board

The Kanban board was updated at the close of iteration 1 to reflect the completed work, and all of the iteration 1 tasks have been moved to the Done column. Lastly, the iteration 2/3 items remain in Backlog for the next sprint.

Backlog	In Progress	In Testing	Done
Algorithm - 10d (P40) Account Deletion - 6d (P10) Game Visuals - 6d (P20)			Game Mechanics - 14d Stats Display - 4d Win/Loss Ratio - 4d Friends List - 4d Repo Setup - 2d App Skeleton - 3d QA Sign-off - 2d

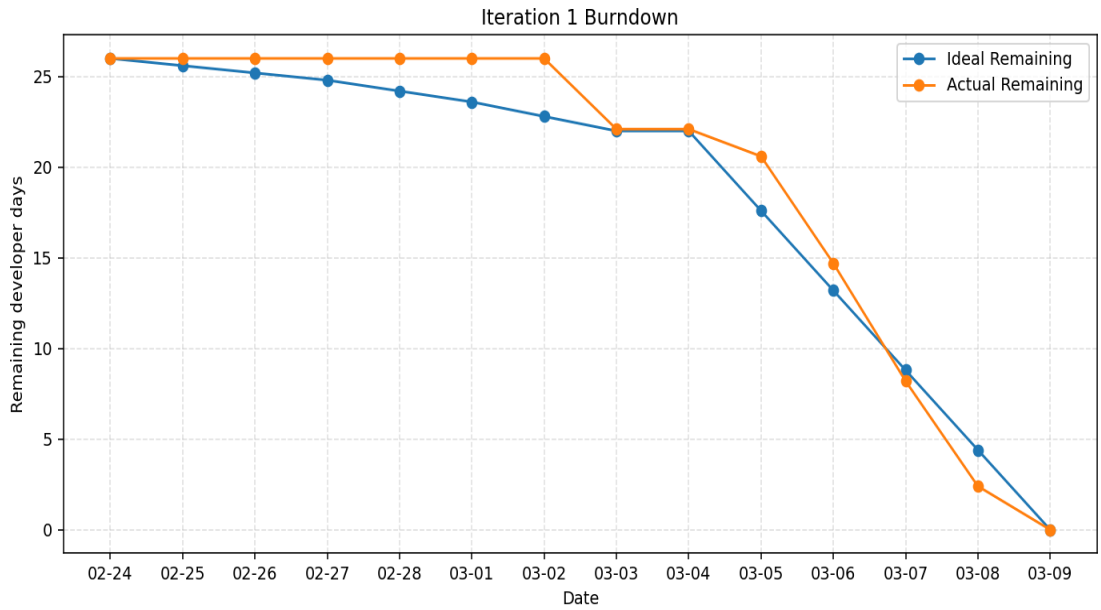
4. Challenges & The Next Steps

4.1 Technical & Process Challenges

Challenge	Fix
The feature estimates were inaccurate	Break the tasks into smaller pieces before assigning
Integration happened too late	Short check ins with each other with frequent pulls from develop to ensure everyone is on the same page
Team communication was inconsistent	More check-ins between the group are scheduled
Unclear task boundaries caused rework	Define the module ownerships earlier

4.2 Velocity & Burndown

All 26 tracked developer days of core implementation work for iteration 1 was completed, and the updated burndown chart below shows the progression of the project during this iteration from day 0 (February 24) to day 13 (March 9). And at the end, the actual remaining work left converged to 0 at the end, this yields a velocity of approximately 26 developer days per iteration (approx. 79% of the full 33 day iteration).



4.3 Goals for Iteration 2

Based on the iteration 1 retrospective and the phase 1 plan, iteration 2 (Weeks 3-4) will focus on:

- **Recommendation Algorithm (10d, P40) -** implementing a game recommendation system based on the user's own play history and friend activity
- **Third Game and Current Game Refinement** - implementing a the third gambling game to meet the minimum 3-game requirement as well as improving the current games before the final submission.
- **Server Side Integration Planning -** plan the connection from the Java WebSocket back end to the front end for real-time updates.

Updated Iteration 1 Task Board

ID	Task	Effort	Assigned To	Status
I1-01	Initial repository setup + branch model	2 dev days	Hayden Dunn	Done
I1-02	App skeleton and module wiring	3 dev days	Hayden Dunn	Done
I1-03	Game mechanics implementation (Slots + Dice)	14 dev days	Nikola Grujin + Adrian Ahmadi	Done
I1-04	Friends list implementation	4 dev days	Mele Felix + Adrian Ahmadi	Done
I1-05	Statistics display implementation	4 dev days	Eric Beaulne	Done
I1-06	Win/loss ratio implementation	4 dev days	Eric Beaulne	Done
I1-07	Integration + QA sign-off	2 dev days	Hayden Dunn	Done
I2-01	Recommendation algorithm	10 dev-days	Nikola Grujin + Mele Felix	Iteration 2
I3-01	Account deletion	6 dev-days	Eric Beaulne + Hayden Dunn	Iteration 3